Transverse Wave Generation on a String

1. Definition

A transverse wave is a wave in which the particles of the medium vibrate perpendicular to the direction of wave propagation.

When a string is vibrated up and down, transverse waves are produced along the string.

2. Principle

Transverse waves are generated on a stretched string by applying a periodic force at one end. The disturbance travels along the string, while the particles of the string vibrate perpendicular to the direction of propagation.

3. Experimental Arrangement

A light string is stretched horizontally and attached to a vibrator at one end. The other end passes over a pulley and supports a hanging weight to maintain tension.

When the vibrator oscillates vertically, transverse waves travel along the string.

4. Formation of Stationary Waves

When the travelling waves are reflected from the fixed end, the incident and reflected waves superpose and may form stationary waves.

Stationary waves consist of:

- Nodes: Points where displacement is always zero.
- Antinodes: Points where displacement is maximum.

For a string fixed at both ends, the ends are nodes.

5. Important Formulae

Wave velocity:

$v = \sqrt{(T/\mu)}$

Where:

- v = velocity of the wave
- T = tension in the string
- μ = mass per unit length of the string

Mass per unit length:

$\mu = m/L$

Where:

- m = mass of the string
- L = length of the string

Wave relation:

$v = f\lambda$

Where:

- f = frequency
- λ = wavelength

Frequency of the nth harmonic for a string fixed at both ends:

$f_n = nv/(2L)$

Where n = 1, 2, 3, ... and L is the vibrating length of the string.

For the fundamental frequency:

$f_1 = v/(2L)$

6. Factors Affecting Wave Velocity

1. Tension: Increasing tension increases wave velocity.
2. Mass per unit length: Increasing mass per unit length decreases wave velocity.
3. Length: The wave velocity is independent of string length when tension and mass per unit length remain constant.

7. Applications

1. Used in understanding the working of stringed musical instruments.
2. Used to study stationary waves and resonance.
3. Used to determine the frequency of vibration of a string.
4. Used in laboratory experiments to verify the laws of vibrating strings.

8. Conclusion

Transverse waves on a stretched string are produced by vibrating the string perpendicular to its length. Their velocity depends on the tension and mass per unit length of the string.

Important formulae:

- Wave velocity: $v = \sqrt{(T/\mu)}$
- Wave relation: $v = f\lambda$
- Fundamental frequency: $f_1 = v/(2L)$